

investing and applied the same type of scientific thinking and mathematical reasoning to each of these endeavors. It is therefore not surprising that the same principles and risk management he employed in gambling are as pertinent to providing trading lessons as those he used in the markets.

Beating roulette was once considered a mathematical impossibility. By approaching the problem in a completely different way—focusing on predicting the most likely termination zone for the ball—Thorp was able to create a giant 44 percent edge. The broad lesson is that sometimes what seems impossible is entirely possible if approached from a completely different perspective. A similar condition might apply to trading. Even if one grants that the markets can't be beat with conventional approaches, it by no means implies that unconventional approaches would have the same limitation.

Thorp's insight in how to improve the odds in blackjack betting vis-à-vis the "basic system" (that is, correct probabilistic decisions on when to hit, hold, split, or double down) was to vary the bet size. By betting more on high-probability hands than on low-probability hands, Thorp was able to transform a losing game—even one played with perfect decisions (other than bet size)—into a game with a significant positive edge. There is an important analogy here that applies to traders: Varying the position size can be as important as the entry methodology. Trading smaller, or not at all, for lower probability trades and larger for higher probability trades can transform a losing strategy into a winning one. (Thorp had to bet something on even low-probability blackjack hands, but the trader can forgo lower probability trades altogether.)

Although, in trading, probabilities cannot be accurately assessed, as they can in blackjack, traders can often still differentiate between higher and lower probability opportunities. For the quantitative trader, such an assessment might be based on a statistical analysis of past results for different strategies. As for discretionary traders, many of them have strong differences in confidence level for different trades. If a trader does better on high-confidence trades, then the degree of confidence can serve as proxy for probability of winning. The implication then becomes to trade larger on high-confidence trades and smaller, or not at all, on low-confidence trades.